

ABEKA SCHOLAR ACADEMY

Subj Grade 5 Science & Biology Curriculum

Official Curriculum & Study Material

Chapter 1

Welcome to our amazing journey into science & biology! We have so many fun things to discover today. Let's get started! Science is what we do to find out about the natural world. There are different kinds of science. Natural sciences study nature and the physical world. They include biology, chemistry, geology, astronomy, and physics. Social sciences study people and how societies work. They include, psychology, sociology, and economics. Applied sciences use the things we learn from science to solve problems. They include, engineering and medicine. Science uses mathematics, computer science and logic, which are sometimes called "formal sciences". Science makes observations and experiments. Science produces correct facts, scientific laws, and theories. 'Science' also refers to the large amount of knowledge that has been found using this process. Research uses the scientific method. Scientific research uses hypotheses based on ideas or earlier knowledge, which can be categorized through different topics. Then those hypotheses are tested by experiments. People who study and research science and try to find out everything about it are called scientists. Scientists study things by looking at them very carefully, by measuring them, and by doing experiments and tests. Scientists try to explain why things act the way they do, and predict what will happen.

This is one of my favorite things to teach! The history of science is thousands of years old. The beginnings of modern science can be found in ancient Egypt and Mesopotamia (around 3000–1200 BCE). They developed early ideas in math, astronomy, and medicine. They influenced the Greeks, who tried to explain what happened in our world using natural causes. Later, during India's Golden Age, more progress was made. This included the creation of the Hindu–Arabic number system. Here's a really cool secret about this topic: After the Western Roman Empire fell during the Early Middle Ages (400–1000 CE), scientific progress in Europe slowed down. But progress started up again during certain periods. In the Islamic world, scholars kept, studied, and made Greek science better during the Islamic Golden Age. Later, when the Byzantine Empire got weaker, its scholars took Greek books to Western Europe, helping start the Renaissance. From the 10th to 13th centuries, Europe re-learned Greek and Islamic science. This helped cause the scientific revolution in the 16th century. Science began to change a lot. New discoveries removed old ideas. The scientific method became more important. By the 19th century, science became more professional and organized. Instead of calling it “natural philosophy”, it was called “natural science”.

The history of the word “science” and the terms used in science tell us a lot about how the subject has

changed over time and what it means in different cultures and philosophies. The word “science” comes from the Latin word “scientia,” which means “knowledge.” That word came from the verb “scire,” which means “to know.” In ancient Rome, scientia just meant any kind of knowledge you could get through learning or experience. It included many subjects like philosophy, speaking (rhetoric), and math. But back then, it did not mean what we now think of as science. In ancient Greece, a similar idea was called “epistēmē” (ἐπιστήμη). Philosophers like Aristotle and Plato used this word to mean knowledge that could be proven through reason. They saw it as different from “doxa,” which meant opinion, something that might not be true or could change. Imagine we are going on an amazing adventure to learn about this! The meaning of “science” started to change during the Scientific Revolution in the 1500s and 1600s. During this time, famous thinkers like Galileo Galilei, Johannes Kepler, Francis Bacon, and Isaac Newton began to create a new way of learning about the world. They used observation, experiments, and math to find answers. This method helped form the basis of what we now call modern science. Back then, people did not call it “science”, they called it “natural philosophy”, especially when studying the physical world like motion, planets, or light. The word “scientist” was not even invented until 1834. A man named William Whewell, who studied the history and philosophy of science, came up with it. He suggested “scientist” in a book review to describe someone who studies the natural world in a careful, organized way, just like the word “artist” describes someone who creates art. Science words (or scientific terms) have changed and grown over time, just like science itself. In the past, many science terms came from Greek and Latin, because those were the main languages used by educated people in Europe for many centuries. For example, the word “biology” comes from the Greek words *bios* (which means “life”) and *logos* (which means “study”), so biology means “the study of life.” The word “physics” comes from the Greek word *physis*, meaning “nature.” “Chemistry” likely comes from the Arabic word *al-kīmīyā* (الكيمياء) which has roots in ancient Egyptian and Greek (Hellenistic) alchemy, a practice that tried to change substances and discover how materials work.

Scientific terms are very important because they help scientists be precise (use exact meanings), communicate clearly, and organize ideas and information. Each area of science has its own set of special words to describe things it studies. For example, in physics, words like “momentum,” “entropy,” and “quantum” have very specific meanings that are different from how they are used in everyday language. In biology, scientists name living things using a system called Linnaean binomial nomenclature. This system gives each organism a Latin name made of two parts: the genus and species. For example, humans are called *Homo sapiens*. Having standardized scientific words is important for making sure experiments can be repeated correctly (replicability), letting scientists check each other’s work (peer review), and helping scientists from all over the world work together,

even if they speak different languages. Here's a really cool secret about this topic: Many words used in science today come from the history and culture of the time when they were first used. Some scientific terms actually started out as jokes, creative ideas, or simple descriptions, but they became official because people started using them so much. For example, the term “Big Bang” was first used by astronomer Fred Hoyle in a radio broadcast in 1949. He did not believe in the Big Bang theory and actually used the name to make fun of it. But, the name caught on and is now the main word we use for how the universe began. Another example is the word “quark,” which is used for tiny particles that make up protons and neutrons. Physicist Murray Gell-Mann chose this word from a line in a book called *Finnegans Wake* by James Joyce. This shows how creativity and language play a role in science. Scientific words often come from metaphors or comparisons to help people understand complex ideas. For example, the word “cell” in biology was first used by Robert Hooke in 1665. When he looked at cork under a microscope, he saw tiny box-like spaces that reminded him of the small rooms (cells) where monks lived. That is how the name “cell” was born. In computer science, the word “virus” is used to describe a program that can copy itself and infect other programs, just like real viruses do in the human body. Scientists used this name because it helped people understand how these harmful programs behave. These examples show that scientific words often come from real-world ideas, and they help us understand new or difficult concepts by connecting them to things we already know.

The language used in science is always changing to become more fair, clear, and inclusive. In the past, many scientific terms were based on ideas that focused too much on humans, Europe, or one gender. Over time, scientists have worked to replace these words with ones that are more neutral and accurate. For example, in astronomy, the word “planet” was officially redefined in 2006 by the International Astronomical Union (IAU). Because of this change, Pluto was no longer called a full planet, but instead a dwarf planet, which caused a lot of debate. In everyday science language, people now often say “human” instead of “man”, and “humankind” instead of “mankind”. These changes help make science feel more welcoming to everyone, no matter their background or identity. Close your eyes and picture this... As science keeps moving forward, new inventions and discoveries also bring new words. For example, scientists discovered CRISPR, a tool that can edit genes. The word CRISPR is an acronym (a short word made from initials) for “Clustered Regularly Interspaced Short Palindromic Repeats.” It shows how scientists create short terms to explain big, complex ideas. New fields like machine learning, artificial intelligence (AI), and quantum computing are also creating new words. These words need to be defined very clearly, so scientists from different areas can understand each other and work together without confusion. Close your eyes and picture this... The period of prehistoric science includes the Palaeolithic, Mesolithic, and Neolithic ages, which lasted from about

2.5 million years ago to around 3000 BCE. Even though these early efforts were not "science" the way we know it today, they showed careful observation, experiments, and the start of organized thinking. One of the first and most important forms of early science was making stone tools. It started with a human ancestor called *Homo habilis* and continued through different prehistoric cultures like the Acheulean, Mousterian, and Upper Paleolithic. People made tools like hand axes, scrapers, awls (used for poking holes), and spear tips. To do this, they had to understand how stones break, what kinds of rocks work best, and how to hit them with the right amount of force. Archaeologists (scientists who study ancient human history) have found evidence that these early humans learned by trial and error, and they shared their knowledge with others. Over time, they improved their tools to make them stronger and more useful. This kind of hands-on learning and passing down of skills is one of the earliest examples of using science to solve problems, especially in understanding materials and how to use them, what we now call materials science.

Chapter 2

Early humans also showed basic knowledge of astronomy, the study of the sky. This can be seen in archaeological discoveries like the Blanchard bone (around 30,000 BCE) and the Lascaux Cave paintings (around 17,000 BCE). These objects and drawings may show that early people were watching the Moon's phases or recognizing star groups like the Pleiades constellation. One of the earliest sky-watching structures is the Nabta Playa stone circle in southern Egypt, built around 6000 BCE. It is one of the oldest known examples of people using megaliths (large stones) to line up with the summer solstice sunrise, the longest day of the year. It may also have helped track seasonal rains, which were very important for survival in that dry area. These structures show that early people spent years carefully watching the sky, noticing patterns, and trying to predict natural cycles, like when rain or seasons would come. This was an early step toward the science of astronomy. Another major scientific step happened during the Neolithic Revolution, around 10,000 BCE, when humans started farming. This time period was extremely important because people learned to grow plants and raise animals. They had to understand soil, water, and weather patterns to grow food successfully. Early farmers in places like the Fertile Crescent (in the Middle East), China's Yellow River valley, and Mesoamerica (Central America) grew crops like wheat, barley, millet, rice, and maize (corn). They watched how these plants grew over many generations and chose the ones with the best traits, like more food, better taste, or the ability to survive without much water. This careful choosing is the beginning of genetics (even though they did not know that word), and their work helped people settle

down, build villages, and eventually start civilizations. These farming skills are the beginning of what we now call agricultural science. Even though prehistoric people did not write things down, they still had surprisingly advanced medical knowledge. One of the most amazing examples is trepanation, a kind of surgery where people removed parts of the skull. Scientists have found skulls from as far back as 7000 BCE in Europe, Africa, and South America that show signs of this surgery, and some of the skulls even show healing, which means people survived the operation. This suggests that early humans had basic knowledge of the body, special tools, and may have used natural medicines, like herbs, to reduce pain or stop infection. Another example is Ötzi the Iceman, a frozen body found in the Alps from around 3300 BCE. His body had tattoos that lined up with places used in acupuncture, a method of healing still used today. He was also carrying a medicinal fungus called *Fomitopsis betulina*, which may have been used to fight infection.

People in prehistoric times also began using math and measurement. They made tools like tally sticks to help them count. One of the oldest is the Lebombo bone from about 35,000 BCE, and another is the Ishango bone from around 20,000 BCE. These bones have carved marks that might show number patterns, multiples, or even prime numbers. Some scientists think they may also have tracked lunar cycles (the phases of the moon). Later, in places like Mesopotamia and the Indus Valley, early people started using standard measurements. They needed these for storing grain, trading goods, and building things. They created systems to measure volume and weight, which helped them with farming, business, and early forms of engineering. In prehistoric times, early people showed amazing skills in building large stone structures, which we now call megalithic architecture. One of the oldest examples is Göbekli Tepe, built around 9600 BCE in what is now Turkey. It has huge stone pillars carved with animals, arranged in circles. These stones are very heavy, and moving and placing them would have taken many people working together, using simple tools and their understanding of leverage (how to move heavy things using force and angles). They probably also used basic measuring tools to plan where to put the stones. Other examples, like Stonehenge in England (about 3000 BCE) and the stone lines at Carnac in France, show that people in different places had similar building knowledge. These projects needed not just strength, but also careful planning, measuring, and even watching the sky to line up stones with seasons or sunrise points. This shows early knowledge of engineering and astronomy. Let me tell you a fascinating story. Early humans were also smart about taking care of the environment. They used fire in helpful ways. For example, Australian Aboriginals practiced something called fire-stick farming. They burned certain areas on purpose to help new plants grow, bring animals to the area, and make the land easier to manage. In the Andes Mountains (in South America), ancient farming communities built terraces (step-like farms on hills) and irrigation systems to control water. This helped them farm on steep land and stop soil from washing away. These people

understood how to manage natural resources, like water and soil, in ways that are similar to today's ideas of sustainability (protecting the environment while meeting human needs).

People also started to create ways to record and share information. Early forms of writing included cave art, petroglyphs (pictures carved in rock), and tokens (small clay objects used for counting). In Mesopotamia, around 8000 BCE, people used clay tokens to keep track of food and goods. Over time, this system turned into writing, known as cuneiform. Writing allowed people to store knowledge, like farm records and star charts, and pass it on to future generations. This step, moving from oral traditions (speaking stories and knowledge) to writing things down, helped humans build on knowledge over time. It was the beginning of being able to record science and discoveries so they would not be forgotten. Ancient science was the early scientific knowledge and inventions created by some of the first civilizations in places like Mesopotamia, Egypt, the Indus Valley, ancient China, and Mesoamerica (Central America). This period lasted from around 3000 BCE to 600 BCE. In Mesopotamia, one of the first places where civilization began, science was used in areas like farming, government, and astronomy. The Sumerians and Babylonians created number systems, including one based on 60 (sexagesimal). This is why we still use 60 minutes in an hour and 360 degrees in a circle today. They were also great at studying the sky. On clay tablets written in cuneiform writing, they recorded things like star charts, calendars, and math problems. Babylonian astronomers carefully watched the Moon and planets, and used math to predict eclipses (when the Moon or Sun is covered) and planet alignments. One of their most important discoveries was the Metonic cycle, which is a 19-year cycle where the Moon's phases happen on the same days of the calendar again. This is one of my favorite things to teach! Ancient Egyptian science was closely tied to farming, timekeeping, and building large monuments. The Egyptians created a 365-day calendar by watching the star Sirius (which they called Sopdet). Every year, Sirius rose in the sky just before the Nile River flooded. This was a very important event that made the soil good for growing crops. To prepare for this, the Egyptians had to be good at measuring the sky and water levels. They also used science to build huge structures like the pyramids. These amazing buildings needed careful planning, geometry (math involving shapes), and knowledge of materials and surveying (measuring land). Egyptian engineers were skilled at all of these things. A famous document called the Rhind Mathematical Papyrus, written around 1650 BCE, shows that Egyptian students learned math such as fractions, areas, volumes, and ratios.

Close your eyes and picture this... The Indus Valley Civilization (around 2600 to 1900 BCE) also had amazing science and technology, even though we still do not fully understand it because their writing system has not been decoded yet. Cities like Mohenjo-Daro and Harappa were built in neat grid

patterns, had advanced drainage systems, and used standard sizes for things like weights and bricks. At a site called Lothal, scientists found a dockyard, which shows that people there understood how tides worked and how to build for the sea. The use of uniform bricks and measuring tools also shows they had organized production and possibly government systems to manage it. All of this points to high-level technical knowledge and careful planning. In ancient China, science grew together with philosophy and government. People studied nature because they wanted to understand it better and use that knowledge to rule wisely. One early book called the I Ching (also called the Book of Changes) was written by at least the late 2nd millennium BCE. It used patterns and binary logic (ideas based on two choices, like yes/no or dark/light) to explain natural cycles and how the universe worked. By the time of the Zhou Dynasty (around 1046–256 BCE), Chinese scientists had invented many useful tools and ideas. They created early earthquake detectors (called seismographs), studied the stars (astronomy), used plants as medicine (herbal medicine), and learned how to work with metals and chemicals (metallurgy). They also made a complex calendar based on both the Moon's phases and solar seasons. This calendar used a 60-part cycle (called the sexagenary cycle) and showed that they had a very advanced way of measuring time using math. The Chinese were also experts in bronze casting, making tools and objects out of melted metal. They made alloys (mixtures of metals) that had the same recipe every time, using careful control of heat and basic understanding of chemistry. In Mesoamerica, the Olmec (around 1500–400 BCE) and later the Maya made amazing discoveries in astronomy, math, and building. The Maya used a base-20 number system (instead of base-10 like we use). They invented long count calendars that could track time for thousands of years. They watched the sky carefully and recorded things like the movements of Venus, eclipses, solstices, and planet alignments. Some of this information was written in books called codices, like the Dresden Codex. The Maya used these observations to plan religious festivals, decide when to plant crops, and even schedule political events. They built pyramids and temples that lined up perfectly with events in the sky. They were constantly watching the sky and collecting data, just like modern scientists.

In all of these ancient civilizations, medicine was based on practical experience and observation, even though it was often connected to religion or spiritual beliefs. In Ancient Egypt, a medical text called the Edwin Smith Papyrus (written around 1600 BCE) showed that Egyptian doctors had a logical and scientific approach to treating injuries. It includes detailed instructions for doing surgery and describes how to treat injuries to the head, spine, and arms or legs. It even gave predictions (called prognosis) about how serious the injuries were and what might happen to the patient, depending on where the injury was. In Mesopotamia, doctors used a book called the Diagnostic Handbook by a man named Esagil-kin-apli (around 1069 BCE). This book helped doctors figure out what was wrong with a patient by looking at their symptoms, then choosing treatments. These treatments might include

herbal medicine, chants or prayers, and physical treatments like bandages or surgeries. In Ancient China, a book called the *Shennong Bencaojing* (written in the first millennium BCE) listed hundreds of medicinal herbs and their effects on the body. This book became the base of Traditional Chinese Medicine. It showed how Chinese doctors tested herbs, classified their properties, and wrote down their results, which helped build knowledge over many generations. These early sciences also depended on good recordkeeping, which allowed people to share and pass on knowledge over time. Each civilization created its own writing system, including cuneiform in Mesopotamia, hieroglyphs in Egypt, oracle bone script in China, and Maya script in Mesoamerica. These writing systems helped people write down important information, like star charts and astronomy, math tables, medical recipes, and instructions for building things. Because this knowledge was written down, it could be passed from one generation to the next, even if science back then was usually done by priests, scribes, or royal advisors, not scientists like we think of today. Classical science started in ancient Greece, especially in a city called Miletus. There, early thinkers like Thales (around 624–546 BCE) tried to explain how the world works without using stories about gods or magic. Thales believed that water was the basic substance that everything came from. This idea started a new way of thinking called natural philosophy, which focused on finding natural causes for things we observe in the world. After Thales, other thinkers from Miletus also had big ideas. Anaximander said everything came from something called the *apeiron* (which means “the boundless” or unlimited). Anaximenes thought air was the most important element.

Chapter 3

Let me tell you a fascinating story. Another important thinker was Pythagoras (around 570–495 BCE). He believed that numbers and math were the true key to understanding the universe. His school studied geometry, music, and numbers. They discovered that when strings of certain lengths were plucked, they made sounds that matched simple number ratios. This showed a connection between music and math, and supported the idea that the whole universe had a kind of mathematical harmony, a concept called the “music of the spheres.” In medicine, a famous Greek named Hippocrates (around 460–370 BCE) made big changes. He said that illness was not caused by angry gods or magic, but by natural causes like diet, environment, and the balance of fluids in the body. He taught doctors to carefully observe patients, keep records, and study the effects of things like climate and food on health. Hippocrates also created the four-humor theory, which said the body had four fluids: blood, phlegm, black bile, and yellow bile. People believed for many centuries that staying healthy meant

keeping these in balance. He also introduced ethical rules for doctors, such as those in the Hippocratic Oath, which is still remembered today. Because of Hippocrates, medicine became more scientific and professional. Aristotle (384–322 BCE) was one of the most important thinkers in ancient science. He was a student of Plato and later became the teacher of Alexander the Great. Aristotle wrote about many subjects, including biology, physics, astronomy, psychology, and logic. In biology, Aristotle studied sea animals and did some of the first dissections (cutting open bodies to learn how they work). He grouped animals by their physical traits, which helped create early classification systems. In physics and astronomy, he believed the Earth was at the center of the universe (geocentric model), and that everything in the sky moved in perfect circles made of a special substance called aether. He thought the heavens were unchanging, unlike Earth, which was made of earth, water, air, and fire, things that could decay and change. Aristotle also created a famous way of thinking about why things happen, called the four causes. The material cause was what something is made of, the formal cause was the shape or structure of the thing, the efficient cause was what caused it to happen, and the final cause was its purpose or goal. One of Aristotle's students, Theophrastus (around 371–287 BCE), took these ideas further, especially in plant science. He wrote books like *Enquiry into Plants* and *On the Causes of Plants*, where he tried to classify and describe plants in a clear and organized way. He noticed details like where plants grow, how they reproduce, and how people use them. He divided plants into groups like trees, shrubs, and herbs, and because of his work, he is known as the “father of botany”. His ideas about plants were so good that scientists still read them over a thousand years later. After Alexander the Great died, Greek science continued to grow during the Hellenistic period (323–31 BCE). One of the most famous centers of learning was Alexandria, in Egypt. It had a huge library, scholars from many cultures, and support from powerful leaders. One of the most important scientists there was Euclid (around 300 BCE). He wrote a book called *The Elements*, which organized everything known about geometry into 13 books. He proved shapes and math rules using logic and step-by-step thinking. This method, called the axiomatic method, became the model for how scientists and mathematicians would work for thousands of years.

Here's a really cool secret about this topic: Archimedes of Syracuse (about 287–212 BCE) was a brilliant ancient scientist and inventor. He made important discoveries in physics, engineering, and math. One of his most famous ideas is called Archimedes' principle. It says that when you put something in water (or another liquid), the liquid pushes up on it with a force equal to the weight of the liquid the object pushes out of the way. This explains why some things float and others sink. Archimedes also studied levers, pulleys, and other machines, helping to create the basic rules of mechanical engineering. He figured out how to calculate the area and volume of different shapes and came up with a very good estimate for the number pi (π). Close your eyes and picture this...

Eratosthenes (about 276–194 BCE) was the head librarian at the Library of Alexandria, one of the greatest learning centers of the ancient world. He was good at math, astronomy, and geography, and used these skills to measure the size of the Earth very accurately. He did this by comparing shadows in two cities at the same time during the summer solstice (the longest day of the year). He also invented a system of latitude and longitude to map locations, just like the grid system used in maps today. Eratosthenes tried to create a map of the entire known world. Aristarchus of Samos (about 310–230 BCE) was an early astronomer who had a bold idea: the Sun is at the center of the solar system, not the Earth. This idea is called the heliocentric model. At the time, most people believed the Earth was at the center, so his theory was not accepted, partly because they did not have strong telescopes or enough evidence. Even though his idea was ignored back then, his work later inspired scientists like Copernicus, who helped start the modern understanding of the solar system. Claudius Ptolemy (about 100–170 CE) lived in Roman Egypt and was a famous ancient astronomer and scientist. He believed that the Earth was at the center of the universe, and he wrote a big book called the *Almagest*, where he explained this idea using a model called the geocentric model. To explain how the Sun, Moon, and planets move in the sky, he used epicycles, deferents, and equants, complicated ways of showing orbits that looked like circles within circles. Even though this model was wrong (we now know the Sun is at the center), Ptolemy's system worked well for making predictions and was used for over 1,400 years. Ptolemy did not just study space, he also worked on optics (how light and vision work), maps, and music. He even wrote about astrology.

Did you know? The Romans were more focused on practical science than on theories. They were great engineers and used science to build things like aqueducts (which carried water), roads, bridges, and concrete buildings. They understood how water flows, how to balance weight, and how to use strong building materials. A Roman engineer named Vitruvius wrote a book called *De Architectura* in the 1st century BCE, where he talked about architecture, human body proportions, and machines. He combined art and science to show how buildings should look beautiful but also work well. In medicine, a Roman doctor named Galen (around 129–200 CE) became very important. He studied how the body works by examining animals, learning about organs, bones, and how diseases spread. He believed in the four humors (like Hippocrates) and thought that keeping these in balance kept people healthy. Galen's medical ideas were used in Europe and the Islamic world for many hundreds of years. Science during the Islamic Golden Age (from about the 8th to the 14th century CE) was one of the most creative and important times in history. During this period, Muslim scholars made huge advances in astronomy, math, medicine, physics, chemistry, geography, and engineering. This progress happened because of several things which included leaders supported science, especially the Abbasid caliphs who helped fund research and learning. Books from Greece, India, Persia, and China

were translated into Arabic, allowing people to learn from many different cultures, and Islamic culture encouraged learning (called ‘ilm in Arabic) and logical thinking. One of the most important places during this time was the House of Wisdom (Bayt al-Hikma) in Baghdad, created by Caliph Harun al-Rashid and expanded by his son Al-Ma'mun. It was a major center where scholars gathered to translate, study, and improve scientific knowledge. They translated famous works from Greek, Persian, and Indian scientists, like Aristotle, Galen, Euclid, and Ptolemy, into Arabic. One key figure was Hunayn ibn Ishaq, who translated over 100 medical books. This is one of my favorite things to teach! In astronomy, Muslim scientists made many improvements to older models from Greece and India. Al-Battani (also known as Albatenius, around 858–929 CE) carefully studied the Sun and calculated the length of the year as 365.2422 days, almost exactly what is known today. Al-Zarqali (also called Arzachel) made the Toledan Tables, which helped European scientists and even influenced Copernicus hundreds of years later. Al-Tusi (1201–1274 CE), working at the Maragheh observatory, invented a math model called the Tusi couple, which helped improve future ideas about how planets move. Muslim astronomers also built observatories (places to study the stars) in cities like Baghdad, Maragheh, Samarkand, and Shiraz. They invented tools like the astrolabe, sextant, and spherical instruments to measure stars and planets very precisely. These tools were used to find the correct times for prayer, determine the direction of Mecca (Qibla), and create Islamic calendars.

Here's a really cool secret about this topic: In the field of mathematics, Islamic scholars made major breakthroughs, especially in algebra, arithmetic, geometry, and trigonometry. One important mathematician, Muhammad ibn Musa al-Khwarizmi (around 780–850 CE), wrote a book called *Al-Kitab al-Mukhtasar fi Hisab al-Jabr wal-Muqabala*. This book laid the foundation for algebra. The word "algebra" actually comes from al-jabr, meaning "completion." Al-Khwarizmi also created step-by-step ways to solve quadratic equations and methods for doing calculations. In fact, the word “algorithm” comes from his name. Another scholar, Omar Khayyam (1048–1131), figured out how to solve cubic equations using geometry. Al-Kashi (died 1429) calculated the number pi to 16 decimal places and improved the way decimal numbers were used, something very important for modern math. In trigonometry, scientists like Al-Biruni and Nasir al-Din al-Tusi made accurate tables for sine, cosine, and tangent, helping to turn trigonometry into its own branch of mathematics. In medicine, Islamic scientists made great progress by using ideas from ancient Greek thinkers and combining them with careful observation and testing. A famous doctor named Al-Razi (also known as Rhazes, 854–925) wrote a huge 25-volume medical book called *Kitab al-Hawi*. He was the first to clearly tell the difference between smallpox and measles and believed in learning through watching and testing patients. Another well-known doctor, Ibn Sina (also called Avicenna, 980–1037), combined medical knowledge from Greece, Persia, and India in his book *The Canon of Medicine*. This book was used as

a guide in both Europe and the Islamic world for more than 600 years. He introduced important medical ideas like testing medicines, using quarantine to stop diseases from spreading, checking a patient's pulse, and understanding how infections work. Muslim doctors also invented surgical tools, used antiseptics to stop infections, and developed special treatments for eye problems. One doctor, Ammar ibn Ali al-Mawsili, even invented a syringe to remove cataracts from the eye. In the study of light and physics, Ibn al-Haytham (also called Alhazen, 965–1040) made discoveries that helped shape modern science. In his book *Book of Optics*, he showed that the Greek idea of vision, where light was thought to come out of the eyes, was wrong. He explained that light bounces off objects and goes into the eye, and he correctly described how the retina and lens work. Ibn al-Haytham tested his ideas with experiments using things like mirrors, lenses, and a camera obscura (an early version of a camera). He believed in forming hypotheses, testing them, and repeating experiments. These are ideas that are important to the modern scientific method. He also studied motion, mechanics, and inertia, and his work influenced later scientists in Europe, like Roger Bacon and Kepler.

Imagine we are going on an amazing adventure to learn about this! In early Islamic times, chemistry was called alchemy, and Muslim scientists made big discoveries while trying to understand how materials change. One of the most important alchemists was Jabir ibn Hayyan (also called Geber), who lived in the 8th century. He wrote hundreds of books about how to heat, purify, and mix substances using methods like distillation and crystallization. He even described how to make aqua regia, a powerful acid that can dissolve gold. Jabir believed in testing ideas through careful experiments and recording results, which helped move alchemy closer to real, practical chemistry. He also used special lab tools like alembics (for distilling liquids), crucibles (for heating substances), and furnaces, helping to shape the way modern chemistry labs would work. Close your eyes and picture this... Muslim scientists also made great progress in geography, which is the study of the Earth and how it is measured. They traveled, observed nature, and made better maps. A brilliant scholar named Al-Biruni (973–1048) used math and trigonometry to measure the size of the Earth from the top of a mountain, and his answer was very close to what we know today. He also collected the exact locations (latitude and longitude) of over 600 cities and even suggested that the Earth spins, hundreds of years before scientists in Europe did. Another scholar, Al-Khwarizmi, corrected mistakes in older maps from the Greek scientist Ptolemy. Later, Al-Idrisi made a famous world map in 1154 for a European king, using knowledge from Arab traders, sailors, and explorers. Islamic scientists also made amazing progress in engineering and building machines. They designed clever devices like early robots (called automata), water clocks, and machines that used water and air to move. In the 9th century, three brothers known as the Banu Musa wrote *The Book of Ingenious Devices*, which described over 100 inventions using gears, valves, and tubes called siphons. Later, a brilliant inventor named Al-Jazari

(1136–1206) built fancy water clocks, mechanical toys that could be programmed to move, and one of the first crankshafts, a tool still used in engines today. These machines were not just for show; they were used in palaces, public fountains, and farming systems to help people in everyday life.

Chapter 4

This is one of my favorite things to teach! To spread knowledge, the Islamic world created many places for learning. These included schools called madrasas, hospitals known as bimaristans, libraries, and observatories where people could study the stars. Famous schools like the Nezamiyeh in Baghdad, Al-Qarawiyyin in Fez (Morocco), and Al-Azhar in Cairo became major centers for science and education. Scholars traveled from place to place, shared ideas, and copied books so they could be studied in different parts of the world. From Spain and North Africa all the way to Central Asia and India, science and learning were an important part of Islamic culture. This is one of my favorite things to teach! During the early Middle Ages (around the 5th to 10th centuries), science in Western Europe slowed down after the fall of the Western Roman Empire. Many ancient books and ideas were at risk of being lost. Luckily, monks in monasteries helped save this knowledge by copying old texts written in Latin. These included the works of famous thinkers like Aristotle, Pliny the Elder, Galen, and Boethius. One important scholar from this time was Isidore of Seville (around 560–636), who tried to collect and organize all human knowledge into one big book called *Etymologiae*. People at the time believed that studying nature helped them better understand God’s creation. Even though science was limited during this period, learning started to improve under Charlemagne (who ruled from 768–814). He supported education and helped start cathedral schools, which taught subjects like math, geometry, music, and astronomy (called the quadrivium), along with grammar, logic, and writing (the trivium). Between the 11th and 13th centuries, science in Europe came back to life thanks to a big effort to translate ancient books. In places like Spain, Sicily, and parts of the Middle East, European scholars came in contact with the Islamic world and the Byzantine Empire. From them, they got access to many old Greek and Roman ideas, now improved by Arabic scholars. Translators like Gerard of Cremona worked hard to turn important Arabic and Greek books by Aristotle, Avicenna, Alhazen, and Galen into Latin so more people in Europe could understand them. These books taught about astronomy, medicine, math, and optics. This sparked the rise of universities in cities like Bologna, Paris, Oxford, and Salamanca. A new way of thinking called scholasticism became popular. Scholastic thinkers like Thomas Aquinas mixed Greek philosophy with Christian beliefs, and others like Albertus Magnus and Roger Bacon encouraged people to observe nature and do experiments to

learn how the world works.

This is one of my favorite things to teach! By the 14th century, some smart thinkers in Europe were starting to question old ideas about how things move. Two of these thinkers, Nicole Oresme and Jean Buridan, worked at the University of Paris and had new ideas about physics. Jean Buridan came up with a theory called impetus, which was an early idea similar to what we now call inertia. He said that once something is moving, it can keep moving on its own. It does not need to be pushed forever. This was a big change from the old belief (by Aristotle) that something had to keep being pushed to keep moving. Nicole Oresme did not agree with the old idea that Earth was the center of the universe (geocentric model). He also used graphs to show how things move, which was an early step toward kinematics, the study of motion. The Renaissance (from the 1300s to the 1600s) was a time when science began to change a lot. People started to focus more on observing nature, doing experiments, and re-reading old science texts from ancient Greece. Scholars like Leonardo Bruni and Marsilio Ficino helped translate these important works so others could study them in more depth. One of the most famous Renaissance thinkers was Leonardo da Vinci (1452–1519). He is known for being a "universal man", or someone who could do many things well. He was an amazing artist, but he also studied anatomy (the human body), how water flows, and how machines work. Leonardo dissected human bodies and made some of the most accurate drawings of muscles, bones, and the heart for his time. He figured out many things about how the vascular system (blood vessels) and body parts worked, even though he never published his findings while he was alive. Imagine we are going on an amazing adventure to learn about this! A big moment in astronomy happened with a scientist named Nicolaus Copernicus (1473–1543). He suggested a heliocentric model of the universe, which means that the Sun is at the center, not the Earth. This idea was published in his book *On the Revolutions of the Heavenly Spheres*. In his model, Earth spins every day and goes around the Sun once a year. At the time, most people believed the Ptolemaic model, which said Earth was the center of everything. Even though Copernicus kept some old ideas like epicycles (small loops in orbits), his model was a huge change and led to a new way of thinking about the universe. Another important astronomer was Tycho Brahe (1546–1601). He did not use a telescope, but he made very careful observations of the stars and planets using special instruments. His data was extremely accurate and helped later scientists understand how the planets move.

Let me tell you a fascinating story. Johannes Kepler (1571–1630), who used Tycho's data, discovered three laws of planetary motion. He showed that planets do not move in perfect circles, but in elliptical orbits (oval-shaped). He also found that planets move faster when they are closer to the Sun. Galileo Galilei (1564–1642) combined experiments and astronomy. He used one of the first telescopes to look

at the sky and made amazing discoveries. He saw craters on the Moon, sunspots, the phases of Venus, and moons orbiting Jupiter. All these things disproved the old idea that everything revolved around Earth and helped support Copernicus's heliocentric model. Galileo also studied how objects fall, how pendulums swing, and how things move through the air. He showed that the laws of nature follow math rules, and we can understand them by doing experiments. In 1632, he wrote a book called *Dialogue Concerning the Two Chief World Systems* that supported heliocentrism. Because of this, he got in trouble with the Catholic Church, which still believed Earth was the center of the universe. Galileo was put on trial and spent the rest of his life under house arrest. Did you know? During the Renaissance, there were huge changes in medicine and anatomy (the study of the human body). A scientist named Andreas Vesalius (1514–1564) made a big breakthrough by dissecting real human bodies and studying them carefully. Before this, people mostly followed the ideas of an ancient doctor named Galen, but Vesalius found that Galen had made many mistakes because he studied animals, not humans. Vesalius wrote a book called *De humani corporis fabrica* (which means *On the Fabric of the Human Body*), where he corrected hundreds of errors and showed that anatomy should be based on direct observation. This helped make anatomy a real science based on what doctors could see and study for themselves. Later, another important doctor named William Harvey (1578–1657) discovered how blood moves through the body. He showed that the heart pumps blood in a circle, a system called circulation, which was very different from the old idea that blood came from the liver and just moved back and forth. Harvey proved this through experiments and live animal dissection (called vivisection). His work helped change medicine from just guessing and theory to something based on evidence and real body functions. In math and physics, the Renaissance also saw big advances. A man named Simon Stevin (1548–1620) came up with decimal fractions, which made it easier to do math with numbers. For example, instead of writing $\frac{1}{10}$ or $\frac{1}{100}$, people could now write 0.1 or 0.01, something used all the time today. René Descartes (1596–1650), a French thinker, invented analytic geometry, which connected algebra with shapes and space. This helped scientists describe the world more clearly using equations. Another important figure was Francis Bacon (1561–1626). He believed science should be based on observation and experiments, not just guessing or using old books. In his book *Novum Organum*, he explained how scientists should use inductive reasoning, starting with observations and then finding general rules. His ideas became the philosophy of science and influenced many future scientists. Finally, Isaac Newton (1642–1727) brought together the work of earlier scientists like Kepler, Galileo, and Descartes. In the early 1700s, he created a complete system of physics and astronomy that explained how planets move and how objects fall on Earth. His work marked the end of the Renaissance and the start of the Scientific Revolution, a time when modern science truly began.

The Enlightenment was greatly influenced by the ideas of the Scientific Revolution, especially the work of Isaac Newton (1642–1727). In 1687, Newton wrote a book called *Philosophiæ Naturalis Principia Mathematica*, where he explained the laws of motion and the law of universal gravitation. He showed that the same force that makes an apple fall also keeps the Moon in orbit around the Earth. Newton also explained that planets move in elliptical (oval-shaped) paths because of gravity. He invented calculus (at the same time as another scientist named Gottfried Leibniz) to help describe things that change, like speed and movement. His experiments with light proved that white light is made up of many colors. Because of Newton, many Enlightenment thinkers started to use math and science to understand more than just space or motion. In chemistry, Robert Boyle (1627–1691) was an important figure. In his book *The Sceptical Chymist* (1661), he rejected old ideas from Aristotle and alchemy and introduced new scientific ideas about matter. He discovered Boyle’s Law, which says that when you make a gas container smaller, the pressure goes up (as long as the temperature stays the same). This showed that gases behave in ways that can be measured and predicted. Boyle also believed that science should be based on careful experiments that could be repeated. The move from alchemy (an early, unscientific form of chemistry) to modern chemistry was led by Antoine Lavoisier (1743–1794), who is often called the “father of modern chemistry.” Lavoisier discovered and named the elements oxygen and hydrogen, and he proved that the old phlogiston theory (which said things burned by releasing a strange substance) was wrong. He showed that in a chemical reaction, the total mass stays the same, which is now known as the law of conservation of mass. Lavoisier also created a clear system for naming chemicals, which helped scientists better organize and classify the elements. In biology and anatomy, scientists during the Enlightenment started to focus more on how living things are organized. Carl Linnaeus (1707–1778) invented the system we still use today for naming living things, called binomial nomenclature. He gave each plant or animal two names, a genus and a species. This helped scientists all over the world talk about the same organisms clearly and consistently. His book, *Systema Naturae*, first published in 1735, listed and organized thousands of plants and animals and became the foundation of modern taxonomy, the science of classification. At the same time, scientists were learning more about how the human body works. Albrecht von Haller studied things like how nerves work, how muscles move, and how the heart beats. Through experiments, he helped people understand how parts of the body function automatically, without us thinking about them. This was a big step forward in the science of physiology.

Close your eyes and picture this... During the Enlightenment, astronomy made big advances thanks to better telescopes and the growth of observatories. Edmond Halley (1656–1742) used Newton’s laws to predict when a certain comet would come back. He was right, and that comet is now named Halley’s Comet. This showed that gravity could explain how objects move in space. Later, William Herschel

(1738–1822) discovered a new planet, Uranus, in 1781. He also found over 2,500 objects in the night sky, like nebulae (clouds of gas and dust) and star clusters. He built some of the biggest telescopes of his time. His work helped scientists realize that the universe is much bigger than just the Milky Way galaxy. Astronomy became more professional, with major observatories in cities like Greenwich, Paris, and Berlin. These places also helped improve navigation, which was important for travel and trade. Scientific institutions were very important during the Enlightenment. The Royal Society of London (founded in 1660) and the French Academy of Sciences (founded in 1666) brought scientists together to share ideas. They published journals, like *Philosophical Transactions*, that explained experiments, new discoveries, and theories. These groups helped make science more organized, trustworthy, and easy to understand by others. They also started the idea of peer review, where other scientists check your work. As more people learned to read and write, science started to spread outside of just schools and universities. Salons, coffeehouses, and public lectures became places where people talked about science. Scientists were no longer just private thinkers, they became public figures who shared their knowledge with everyone. During the Enlightenment, medicine improved because doctors started focusing more on observing patients and learning through hospital training. One important doctor, Giovanni Battista Morgagni (1682–1771), helped create modern diagnosis by showing that symptoms were linked to specific organs, not just to old ideas like imbalances of body fluids. This helped doctors understand that diseases could be traced to problems in certain parts of the body. Another key figure was Edward Jenner (1749–1823), who made the first successful vaccine. He noticed that people who got cowpox did not catch smallpox, a deadly disease. He tested this idea and proved that cowpox could protect people. In 1798, he published his results, starting the science of immunology and vaccination.

Chapter 5

During this time, science became more separate from religion and was seen as a way to improve society. Famous thinkers like Denis Diderot, Voltaire, and Jean le Rond d’Alembert believed that science and reason could free people’s minds and make the world better. They worked on a big project called the *Encyclopédie* (1751–1772), which was like an early encyclopedia. It collected all known scientific and technical knowledge into one place and made it easier for people to learn about how things work, from optics and anatomy to mining and tools. Engineering and technology also made huge progress. James Watt made important changes to the steam engine in the 1760s and 1770s by adding a separate condenser, which made the engine much more efficient. His improvements helped

power factories and machines, leading to the Industrial Revolution. This is one of my favorite things to teach! During the 1800s, science made big changes, especially in how we understand heat, energy, electricity, and atoms. In the study of heat and energy, thermodynamics, scientists created the laws of thermodynamics. These laws helped people understand how steam engines work and how to make them more efficient. In 1824, a scientist named Sadi Carnot came up with the idea of the Carnot cycle, which showed that an engine's efficiency depends on the temperature difference between where the heat comes from and where it goes. This idea helped lead to the second law of thermodynamics, which explains how energy spreads out. Later, Rudolf Clausius introduced the idea of entropy, which measures how energy becomes more disordered over time. Scientists William Thomson (Lord Kelvin) and James Prescott Joule created the first law of thermodynamics, which says that energy cannot be created or destroyed, only changed from one form to another. This is one of my favorite things to teach! In electricity and magnetism, scientists discovered that these two forces are connected. Michael Faraday, in the 1830s and 1840s, did experiments that showed how moving magnets can create electricity, a process called electromagnetic induction. This discovery is what makes electric generators and transformers work. Faraday also came up with the idea of fields, which are invisible areas where electric or magnetic forces can act. Later, in the 1860s, James Clerk Maxwell built on Faraday's work and wrote four equations, called Maxwell's equations, that explain how electric and magnetic fields move and interact. These equations also predicted the existence of electromagnetic waves, which led to the invention of things like radios, TVs, and wireless communication.

At the beginning of the century, John Dalton brought back the idea of atomic theory. He said that every element (like oxygen or iron) is made up of its own kind of atom, and these atoms combine in fixed amounts to make compounds (like water or carbon dioxide). His ideas helped explain important chemistry rules, like the law of multiple proportions, which says that elements always combine in certain number patterns. Later, in 1869, Dmitri Mendeleev created the periodic table. He arranged the elements in order by atomic mass and grouped them by chemical properties. Mendeleev's table was so smart that it even predicted new elements, like gallium and germanium, before they were discovered. When scientists later found these elements and they matched his predictions, it showed that the periodic law was a powerful way to understand and organize chemistry. Did you know? Meanwhile, biology was also changing in a major way. In 1859, Charles Darwin published a book called *On the Origin of Species*. In it, he explained his theory of evolution by natural selection. Darwin said that living things change over time. These changes happen because of small differences that are passed from parents to offspring. If a trait helps an organism survive and have more babies, that trait becomes more common over generations. This is how new species can form. Darwin got many of his ideas by observing animals and plants during his long trip on the ship HMS Beagle. He was also influenced by

Thomas Malthus, who wrote about how populations grow, and Charles Lyell, who studied how the Earth changes slowly over time. Darwin's theory replaced older ideas that species never change. It gave scientists a new way to understand life, one that was based on evidence, change, and history. It also started many important discussions about science, religion, and philosophy. This is one of my favorite things to teach! At the same time as other big discoveries in the 1800s, scientists also developed the cell theory, which says that all living things are made of cells, and that the cell is the basic unit of life. This idea was first introduced in the 1830s by Matthias Schleiden and Theodor Schwann. Later, Rudolf Virchow added that all cells come from other cells, which he summed up in the Latin phrase "omnis cellula e cellula". The invention and improvement of the light microscope helped scientists look more closely at cells. They were able to discover important parts inside cells, like the nucleus, mitochondria, and other tiny structures called organelles. These discoveries helped build the foundation for molecular biology and genetics, which study how cells work and how traits are passed down.

In medicine, the 1800s were a turning point as it began to become more scientific. This progress was made thanks to new knowledge in disease research, germs, surgery, and public health. One of the most important scientists was Louis Pasteur. He showed that tiny organisms (microbes) can cause disease and spoil food. This idea became known as the germ theory of disease. Pasteur proved that germs do not just appear out of nowhere (a belief called spontaneous generation). He also invented vaccines for rabies and anthrax, and created pasteurization, a method to heat food and drinks to kill harmful microbes and keep them from spoiling. Another important scientist was Robert Koch. He discovered the specific bacteria that cause tuberculosis, cholera, and anthrax. Koch also created new lab methods, like using agar plates to grow bacteria and staining techniques to make microbes easier to see under a microscope. He developed Koch's postulates, a set of rules that scientists use to prove that a certain germ causes a specific disease. In the 1840s, doctors began using anesthesia, like ether and chloroform, to perform pain-free surgeries. This was a huge breakthrough because it allowed surgeons to do more complex operations without hurting the patient. Around the same time, Joseph Lister started using carbolic acid (also called phenol) to clean wounds and tools, which greatly reduced the number of people who died from infections after surgery. His work helped start the use of antiseptic techniques in hospitals and led to the modern idea of clean, germ-free medical care. Because of these changes, hospitals became places of real healing and science, not just places where the sick were kept. At the same time, big cities were facing many diseases due to crowding and pollution from the Industrial Revolution. This led to early public health reforms. One important figure was Florence Nightingale, who worked during the Crimean War. She pushed for clean hospitals, better nursing, and proper training, which helped improve medical care all over the world. In the study of geology,

scientists began to understand that Earth is very old, much older than people once believed. Charles Lyell wrote a famous book called *Principles of Geology* (1830–1833), where he explained that natural processes like erosion (wearing down of land) and sedimentation (building up of layers) happen slowly and steadily over a long time. This idea is called uniformitarianism. Lyell's work was important because it challenged the biblical view that Earth was only a few thousand years old. It also gave Charles Darwin the long time span needed for his theory of evolution to work. Other areas of geology, like stratigraphy (studying rock layers) and fossil correlation (matching fossils to different layers), helped scientists piece together Earth's history. Studies in mineralogy (minerals) and paleontology (fossils) also helped explain what the Earth is made of and what life used to be like.

During the 19th century, people began to study human behavior and society in a more scientific way. This was the beginning of the social sciences. A thinker named Auguste Comte came up with the idea of positivism, which said that we should study society using facts and observations, just like scientists study nature. This led to the creation of sociology, a field that tries to understand how things like cities, factories, and social classes affect people's lives. In economics, early ideas from Adam Smith were built upon by other thinkers like David Ricardo, Thomas Malthus, and John Stuart Mill. They used math and statistics to study how money, jobs, and resources are shared in society. This is one of my favorite things to teach! At the same time, science became more organized in every area. Countries created national science academies and professional journals where scientists could publish and share their discoveries. Universities and research institutes trained new scientists, and governments or private groups started to fund science projects. Science also became something the public could enjoy and learn about. Museums and big events like the Great Exhibition of 1851 in London showed off new inventions and discoveries, proving that science was important for progress. Some amazing inventions of the time included the telegraph (for long-distance communication), the steamship, the electric light, and photography. These changes showed how science could improve daily life and move society forward. At the start of the 20th century, scientists realized that the old ideas of physics (called classical mechanics) could not fully explain how very small things (like atoms) or very large things (like galaxies) work. This led to the birth of modern physics. In 1905, a young scientist named Albert Einstein wrote four important papers that changed science forever. One of these introduced the special theory of relativity, which showed that time and space are not fixed. Instead, time can slow down and lengths can shrink when objects move very fast, close to the speed of light. He also came up with the famous equation $E = mc^2$, showing that mass and energy are two forms of the same thing. Later, in 1915, Einstein created the general theory of relativity, which explained gravity in a new way. Instead of thinking of gravity as a pulling force, he said that mass bends spacetime, and this bending is what is felt as gravity. This idea helped scientists understand

things like black holes, gravitational lensing (how light bends around stars), and how the universe is expanding.

Here's a really cool secret about this topic: At the same time, a different kind of physics called quantum mechanics was being developed to understand how things work at the level of atoms and tiny particles. In 1900, Max Planck said that energy comes in small amounts, called quanta. Niels Bohr created a model of the atom where electrons move in specific orbits, helping to explain how atoms give off light. Werner Heisenberg and Bohr developed the Copenhagen interpretation, which included the uncertainty principle. This says we cannot know exactly where a particle is and how fast it is moving at the same time. Erwin Schrödinger came up with wave mechanics, which showed how particles can also act like waves. Paul Dirac combined quantum theory with Einstein's ideas and predicted the existence of antimatter. These discoveries helped create many modern technologies, like semiconductors (used in computers), lasers, nuclear power, and even the new field of quantum computing. In the 20th century, a new area of science called nuclear physics became very important and had a big impact on the world. It started when Henri Becquerel discovered radioactivity, and Marie and Pierre Curie studied it further. They learned how atoms can break apart in a process called atomic decay. In 1938, scientists Otto Hahn and Fritz Strassmann discovered something called nuclear fission, which is when the nucleus of an atom splits, releasing a huge amount of energy. Lise Meitner and Otto Frisch explained how it worked. This discovery led to the creation of the Manhattan Project, a secret program where scientists like Enrico Fermi and J. Robert Oppenheimer developed the first atomic bomb, which was used in World War II in 1945. But nuclear physics was not just used for weapons. It also led to the development of nuclear power plants, which generate electricity, and medical technologies like PET scans (used to see inside the body) and radiation therapy (used to treat cancer). Meanwhile, in biology, scientists began to understand how traits are passed from parents to children. This started with the rediscovery of Gregor Mendel's laws of inheritance, which led to the science of genetics. In 1953, James Watson and Francis Crick, using X-ray images from Rosalind Franklin and Maurice Wilkins, discovered the double helix structure of DNA. This showed how genetic information is stored in the sequence of nucleotides (the building blocks of DNA). This discovery led to molecular biology, which helped scientists understand how genes make proteins and how certain diseases are caused by genetic problems. Later, a huge international project called the Human Genome Project finished in 2003. Scientists mapped out all the DNA in a human, which includes over 3 billion base pairs and more than 20,000 genes. This started the field of genomics, and helped create new ways to study and treat diseases through personalized medicine and bioinformatics (the use of computers in biology).

Chapter 6

Biotechnology in the late 1900s and early 2000s gave scientists amazing new tools to work with DNA and genes. These included Recombinant DNA technology which lets scientists combine DNA from different sources. PCR (polymerase chain reaction) is a method to quickly make millions of copies of a piece of DNA. CRISPR-Cas9, on the other hand, is a tool that lets scientists edit genes very precisely, like using scissors to cut and change DNA. These tools are now used in many ways. To create genetically modified crops, make medicines like insulin, perform gene therapy to treat diseases, and do quick and accurate disease testing. In medicine, a huge breakthrough came in 1928, when Alexander Fleming discovered penicillin, the first antibiotic. This completely changed how bacterial infections were treated and saved millions of lives. Later, vaccines were developed for deadly diseases like polio, measles, and HPV. More recently, mRNA vaccines (like some COVID-19 vaccines) have helped fight new diseases and protect public health around the world. This is one of my favorite things to teach! At the same time, information technology and computer science became a big part of science and everyday life. During World War II, the first electronic computers, ENIAC and Colossus, were built. Alan Turing created early ideas about how computers work and also helped crack secret codes during the war. In 1947, three scientists at Bell Labs, Bardeen, Brattain, and Shockley, invented the transistor, a tiny part that made it possible to build smaller and faster electronic devices. This led to the microprocessor and started the digital revolution. Later, scientists created the internet, which began as a government project called ARPANET. The internet grew into a huge, worldwide system that allows people to communicate, learn, shop, and share information. Space science made huge leaps forward in the 20th century as humans began to explore beyond Earth for the first time. In 1957, the Soviet Union launched Sputnik 1, the first artificial satellite. Then, in 1961, Yuri Gagarin became the first human in space. One of the biggest milestones came in 1969, when NASA's Apollo Program sent Neil Armstrong and Buzz Aldrin to the Moon, making them the first people to walk on the Moon. Robots also helped us explore space. The Voyager probes, launched in 1977, sent back amazing information about the outer planets and are still traveling through interstellar space (the space between stars). In 1990, the Hubble Space Telescope was launched into orbit. It took detailed pictures of galaxies, nebulae, and supernovae, helping scientists learn more about the expanding universe. In the 21st century, space exploration has kept growing. NASA's Mars rovers explore the Red Planet. Space telescopes like Kepler and TESS search for exoplanets (planets around other stars). Private companies like SpaceX and Blue Origin are working on things like Mars missions and even space tourism.

At the same time, climate and environmental science became very important. Back in 1896, scientist Svante Arrhenius said that carbon dioxide (CO₂) in the air could raise Earth's temperature. This is called the greenhouse effect. By the late 1900s, scientists had strong evidence that human activities, like burning fossil fuels, were changing the climate. They got this data from things like ice cores, satellites, and ocean studies. Today, scientists around the world work together through groups like the Intergovernmental Panel on Climate Change (IPCC) to study climate change and warn about its dangers. They say it is urgent to reduce CO₂ emissions to protect the planet. To help with this, we now use tools like climate models, remote sensing (using satellites to study Earth), and renewable energy (like solar and wind power) to fight climate change and find better ways to live sustainably.

Neuroscience and cognitive science are the sciences that study the brain and how we think, feel, and make decisions. In recent years, scientists have made big discoveries thanks to new brain scanning tools like fMRI and PET scans, which show real-time images of brain activity. These tools helped researchers find out that the prefrontal cortex (the front part of the brain) is important for decision-making, and the amygdala (a small part deep in the brain) helps process emotions like fear and anger. Today, scientists are trying to map all the connections in the brain. This is called connectomics. Also, brain-computer interfaces have been developed. These allow people who are paralyzed to move robotic arms or prosthetic limbs using only their brain signals. At the same time, fields like artificial intelligence (AI) and machine learning, which used to be mostly theory, are now real tools. They help with things like voice assistants, face recognition, medical diagnoses, and self-driving cars. This is one of my favorite things to teach!

In the 21st century, scientists from different fields began working together more closely, creating powerful new areas of research. These include nanotechnology (the science of building tiny machines and materials at the scale of atoms and molecules), synthetic biology (designing and creating new living systems, such as bacteria that produce medicine), quantum information science (using the strange laws of quantum physics to make ultra-powerful computers), and systems biology (studying how all the parts of a living system work together). One of the biggest science projects of this century was at CERN, where scientists used the Large Hadron Collider (LHC), a huge machine that smashes particles together, to discover the Higgs boson in 2012. This particle helps explain how other particles get mass and is a key part of the Standard Model of physics. Scientists are also still trying to solve big mysteries like dark matter, dark energy, and how to unite quantum physics with gravity, which could lead to a unified theory of everything in the universe.

Natural sciences are a major area of science that focuses on understanding the natural world. Scientists in this field study how things in nature work by watching carefully, doing experiments, and creating models or theories. These models help explain why things happen and can even predict what might happen next. Natural sciences depend on evidence that can be seen, measured, and repeated.

Scientists use the scientific method, which means they make guesses (called hypotheses), test them through experiments, and change their ideas if the results do not match. This process helps make sure that scientific conclusions are as accurate as possible. There are several main areas of natural science, including physics, chemistry, biology, earth sciences, and astronomy. Each area focuses on different parts of nature, but they often work together in today's research. Physics is the study of matter, energy, and the forces that affect them. Physicists explore everything from the tiniest particles, like electrons and quarks, to giant objects like stars and galaxies. They use different theories depending on the scale. Quantum mechanics for tiny particles and general relativity for huge cosmic objects. Physics has led to important laws and ideas that help explain the universe, such as Newton's laws of motion, the laws of thermodynamics, Maxwell's equations for electricity and magnetism, and the Standard Model, which organizes the basic building blocks of matter. Chemistry is often called the central science because it connects physics and biology. It focuses on what matter is made of, how it is structured, how it behaves, and how it changes during reactions. Chemists study atoms and molecules to understand how they bond together, break apart, and move energy during chemical changes. Chemistry is divided into different areas. Organic chemistry studies compounds that contain carbon (like living things do). Inorganic chemistry looks at compounds without carbon. Physical chemistry explores how chemistry and physics work together. Analytical chemistry focuses on measuring and identifying substances. Biochemistry applies chemistry to biological systems like cells and DNA. Because of chemistry, there are drugs, plastics, batteries, fertilizers, and many other useful materials and tools.

This is one of my favorite things to teach! Biology is the science of living things. It explores how organisms are built, how they work, how they grow, where they come from, and how they change over time. Biology includes many branches, such as cell biology (which studies the basic units of life, cells), genetics (which looks at heredity and DNA), ecology (which studies how organisms interact with each other and the environment), evolutionary biology (which looks at how life has changed over millions of years), and molecular biology (which focuses on life at the smallest scales, like proteins and genes). Biologists use tools like microscopes, DNA sequencing, and gene editing to explore life. Biology is important in health, farming, nature protection, and biotechnology, helping us make vaccines, improve crops, and protect endangered species. Let me tell you a fascinating story. Earth sciences, also called geosciences, are the sciences that study the Earth and everything that affects it. This includes the solid ground, the atmosphere (air and weather), the hydrosphere (water and oceans), and the biosphere (all living things). Earth scientists use ideas from physics, chemistry, and biology to understand how Earth's systems work together. Earth sciences include geology, which looks at the layers of the Earth, rocks, and the history of the planet; meteorology, which studies weather and the

atmosphere; oceanography, which focuses on oceans and sea life; and climatology, which examines long-term climate patterns and how they change. These studies are important for learning about natural disasters like earthquakes, volcanoes, and hurricanes, and for helping us manage natural resources and protect the environment. Close your eyes and picture this... Astronomy is the science that studies space, planets, stars, galaxies, and the universe. Astronomers observe the sky using telescopes and use theories and math to understand what they see. Modern astronomy is closely connected to physics, especially in a branch called astrophysics, which uses physical laws to explain how space objects behave. Astronomy has helped us discover just how huge the universe is and that there are other solar systems beyond our own. Scientists study amazing objects in space like black holes, neutron stars, and dark matter, a mysterious substance that we cannot see directly. Astronomy also helps us explore big questions, like where the universe came from, how it has changed over time, and what might happen to it in the future.

This is one of my favorite things to teach! The natural sciences all use similar methods to learn about the world. First, scientists use empirical observation, which means they carefully collect data using tools or by studying things in nature. They also do experiments, where they test ideas in controlled settings to see what happens. Another method is mathematical modeling, where scientists use math and computers to describe and predict how things work. To make sure results are trustworthy, scientists follow rules like peer review (where other experts check their work), reproducibility (so others can repeat the experiment), and falsifiability (the idea that scientific claims must be testable and could be proven wrong). Scientists use numbers and statistics to support their conclusions with solid, unbiased evidence. Imagine we are going on an amazing adventure to learn about this! Today, the natural sciences often combine different subjects to solve big problems. These mixed areas are called interdisciplinary fields, like biophysics (biology + physics), astrochemistry (astronomy + chemistry), geobiology (geology + biology), and climate science (which combines many fields). These new fields help scientists understand complex challenges, such as climate change, disease outbreaks, clean energy, and space exploration. For example, studying the climate means using knowledge from physics, chemistry, biology, and computer modeling. Scientists in astrobiology use ideas from space science, chemistry, and biology to search for life on other planets. The natural sciences have led to amazing technology and discoveries that have changed our lives. Physics helped create nuclear energy, computers, and electronics. Chemistry made new medicines, better farming products, and useful materials like plastics. Biology and medicine led to vaccines, treatments for genetic diseases, and ways to grow or repair body parts. Earth science helps us predict natural disasters and take care of the environment. Astronomy has led to better telescopes, cameras, and even new ways to process big data. Finally, the natural sciences are very important for education, government, and solving world

problems. Understanding science helps people make smart choices about things like global warming, health, and new technology. Governments use science to make rules, spend money wisely, and protect people and nature. Many big science projects, like CERN (which studies tiny particles), the IPCC (which studies climate change), or the Human Genome Project (which mapped all human genes), show how countries and scientists work together around the world to learn and make progress.

Chapter 7

Empiricism is a basic idea in science. It says that we learn things by using our senses, like seeing, hearing, or measuring, not just by guessing or following tradition. Instead of relying only on what people believe or think is true, science depends on what we can actually observe, test, and measure. This is why scientists use special tools like telescopes, microscopes, and sensors, to help them see things that are too small, too far, or otherwise invisible to the human eye. For example, when scientists look for planets outside our solar system, called exoplanets, they cannot see the planets directly. Instead, they look at tiny changes in the brightness of a star. If the star gets a little dimmer at regular times, it might mean a planet is passing in front of it. This method is based on careful observation, which is what empiricism is all about. Empiricism also means that scientific results should be reproducible. This means that if someone does an experiment again, they should get the same results. A good example is Gregor Mendel's pea plant experiments, where he discovered how traits are passed down from parents to offspring. Scientists were able to repeat his work and get the same patterns, showing that the results were reliable. The idea of empiricism was supported by philosophers like John Locke, who believed that people are not born with knowledge. Instead, he said the mind starts out like a blank slate, and everything we learn comes from experience. This idea is different from rationalism, which says we are born with certain ideas or that we can learn everything through logic alone. In science, empiricism helps make sure our knowledge is based on real-world evidence. Imagine we are going on an amazing adventure to learn about this! Empirical methods in science are not just about watching things happen, they also include doing careful experiments to test ideas. These experiments are done in a controlled way to help scientists figure out which things cause certain results. For example, when Alexander Fleming saw that a certain type of mold killed bacteria, it was not enough just to notice it. Scientists had to do more experiments to prove that the mold, later called *Penicillium notatum*, could really fight infections. This led to the discovery of penicillin, an important medicine. This kind of testing is how scientists confirm what works and what does not. Galileo also used experiments with ramps (inclined planes) to show that objects do not need a force to keep

moving, which went against the old ideas from Aristotle. In more recent times, scientists used special detectors at LIGO to find gravitational waves, tiny ripples in space predicted by Einstein's theory of general relativity 100 years earlier. This discovery was another example of how scientists use careful observation and testing to confirm or disprove scientific theories. Empirical methods are also used in big, complex areas like climate science. Scientists collect information from things like ice cores, satellites, and even tree rings to learn about Earth's climate over time. These different sources of data help build models to understand and predict how the planet is changing.

Even though empiricism, the idea that knowledge comes from observation and experience, is a key part of science, it is not always simple. One challenge is deciding what counts as a valid observation and how to understand what we see. Sometimes, what scientists notice or pay attention to depends on the theories they already believe in. This idea is called "theory-laden observation." Philosopher Thomas Kuhn explained that scientists often use models or ideas they already have to decide what to look for and how to explain it. For example, early astronomers had trouble understanding why planets sometimes seemed to move backward in the sky (called retrograde motion) until they used models like the Ptolemaic system or the later Copernican model, which helped them interpret what they were seeing. In some parts of science, such as quantum physics, things get even harder. Scientists cannot always observe tiny particles directly. Instead, they have to use mathematical models and experiments to figure out what is happening. Phenomena like quantum entanglement or wavefunction collapse are known through indirect evidence and complex data. In the social sciences (like psychology or sociology), empiricism works a bit differently. Scientists use tools like surveys, interviews, and experiments with people to gather data. But this kind of research can have its own problems. For example, people might give answers they think are expected (this is called social desirability bias), or they might act differently because they know they are being watched (called the Hawthorne effect or observer bias). Rationality and logic are very important in science. They help scientists make sure their ideas make sense, their theories fit together, and their conclusions are based on good reasoning. Rational thinking in science means using reason carefully. Scientists use it to come up with ideas (called hypotheses), to plan experiments, to understand results, and to compare different explanations. This helps make sure that conclusions come from real evidence, not just guesses. Logic gives rules for how to go from one idea to another. It helps scientists check if their arguments are valid. For example, in biology, if all mammals are warm-blooded and whales are mammals, then whales must be warm-blooded. This is a basic kind of logical thinking called deductive reasoning. Science also uses the rule of non-contradiction. This means something cannot be both true and not true at the same time. For example, a chemical cannot both contain and not contain the same element under the same conditions. This helps scientists make clear and reliable chemical models. There are also formal types of logic,

like propositional logic, predicate logic, and modal logic. These are used in areas like computer science, artificial intelligence, and mathematical biology. They help scientists build computer programs, design experiments, and understand complex systems. Here's a really cool secret about this topic: Logic and rational thinking are important in science not just to support good ideas, but also to find mistakes, hidden problems, or wrong assumptions in scientific theories. In physics, the discovery of quantum mechanics made scientists question old ideas. For example, classical physics said that objects behave in clear, predictable ways (called determinism) and that things far apart cannot affect each other instantly (called locality). But quantum mechanics showed that very small particles often behave in strange and random ways, so scientists had to use new kinds of logic to understand them. There is also a big problem between two important theories: general relativity and quantum field theory. Relativity explains how big things like planets and galaxies work, while quantum theory explains tiny particles. But the two theories do not fully agree with each other. This has led scientists to search for a new theory that combines both, such as string theory or loop quantum gravity. Science often uses a method called hypothetico-deductive reasoning. This means scientists start with a theory, then use logic to make predictions that can be tested. If experiments agree with the prediction, the theory is stronger. If not, the theory may need to change. For example, we know that antibiotics kill bacteria, not viruses. This logical idea helps doctors avoid giving antibiotics for viral infections, which helps prevent antibiotic resistance. In social sciences, logic is also important. Game theory is a tool that uses logic and reasoning to study how people make choices in situations where others are also making decisions. It helps explain things like voting, diplomacy between countries, and even how governments sell licenses for things like radio frequencies. Game theory helps people make better decisions by thinking logically about the actions of others.

Scientific thinking also includes the idea of parsimony, which is often called Occam's Razor. This idea says that if there are different explanations for something, the simplest one that still explains the facts should be chosen. This rule helps scientists choose the best theories in many fields, such as biology and climate science. For example, a long time ago, people believed that the Earth was at the center of the universe (the geocentric model). But this model needed many complicated parts, like epicycles, to explain how planets moved. Later, Copernicus suggested that the Sun was at the center instead (the heliocentric model). This simpler idea explained the same facts more easily, so it became the better choice. In brain science, a new idea called the Bayesian brain hypothesis says that the brain works by using probability to update what it believes. This means the brain uses logic and evidence, like a smart guesser. This idea is also used in computer fields like robotics and machine learning. Logic is also very important in doing science in a fair and careful way. For example, decision theory is a part of logic that helps scientists decide how to run medical studies. It helps them think clearly about risks,

benefits, and what is unknown when testing new treatments. Today, "science" usually refers to a way of pursuing knowledge, not just the knowledge itself. It is mainly about the phenomena of the material world. The Greek works into Western Europe from the 6th to 7th century B.C. revived "Philosophy". In the 17th and 18th centuries scientists increasingly sought to formulate knowledge in terms of laws of nature such as Newton's laws of motion. And during the 19th century, the word "science" became more and more associated with the scientific method itself. It was seen as a way to study the natural world, including physics, chemistry, geology and biology. Did you know? It was also in the 19th century that the term scientist was created by William Whewell. He meant it to tell the difference between those who looked for knowledge on nature from those who looked for other types of knowledge.

The scientific method is the name given to the methods used by scientists to find knowledge. The main features of the scientific method are: Let me tell you a fascinating story. Scientists identify a question or a problem about nature. Some problems are simple, such as "how many legs do flies have?" and some are very deep, such as "why do objects fall to the ground?" Next, scientists investigate the problem. They work at it, and collect facts. Sometimes all it takes is to look carefully.

This is one of my favorite things to teach! Some questions cannot be answered directly. Then scientists suggest ideas, and test them out. They do experiments and collect data. Did you know? Eventually, they find what they think is a good answer to the problem. Then they tell people about it. Later, other scientists may agree or not agree. They may suggest another answer. They may do more experiments. Anything in science might be revised if we find out the previous solution was not good enough.

Chapter 8

This is one of my favorite things to teach! A famous example of science in action was the expedition led by Arthur Eddington to Principe Island in Africa in 1919. He went there to record where the stars were around the Sun during a solar eclipse. The observation of where the stars were shown that the apparent star positions close to the Sun were changed. In effect, the light passing the Sun was pulled towards the Sun by gravitation. This confirmed predictions of gravitational lensing made by Albert Einstein in the general theory of relativity, published in 1915. Eddington's observations were considered to be the first solid proof in favour of Einstein's theory. Discoveries in fundamental science can be world-changing. For example: The philosophy of science is a subject that looks closely at how

science works and what it really means. It asks big questions like: How do we know something is true in science? How do scientific ideas change over time? Can science truly be objective? People who study the philosophy of science try to understand the methods scientists use, the rules they follow, and how their discoveries affect the world. This field connects to other areas of philosophy, such as epistemology (the study of knowledge), metaphysics (the study of what exists), ethics, and logic. It also looks at how science is actually done in both natural sciences (like physics and biology) and social sciences (like psychology and sociology).

One of the biggest questions in the philosophy of science is called the demarcation problem. This means trying to figure out what counts as real science and what does not, like pseudoscience, which pretends to be scientific but doesn't follow the same rules. A famous philosopher named Karl Popper said that real science should be falsifiable. This means a scientific idea must make clear predictions that could be proven wrong with evidence. For example, Einstein's theory of general relativity predicted that light would bend during a solar eclipse. Scientists tested this and confirmed it, showing the theory was falsifiable. On the other hand, Popper criticized things like Freud's psychoanalysis, which tried to explain human behavior in ways that could not really be tested or proven wrong, because it could twist its explanations to fit any situation. Some later philosophers, like Imre Lakatos and Paul Feyerabend, said that science is not always so simple. They pointed out that many great scientific ideas started out as vague or hard to test. So, the question of what counts as real science is still debated, and there may not be one easy answer. Scientific realism and anti-realism are two different ways of thinking about what science is really doing. People who support scientific realism, like Hilary Putnam and Richard Boyd, believe that scientific theories describe the world as it actually is, even the parts we cannot see, like electrons or black holes. They argue that science works so well, like making accurate predictions and helping us build amazing technology, because its ideas are mostly true, or at least close to the truth. On the other side, anti-realists, like Bas van Fraassen, believe something different. They say the job of science is not to find the absolute truth, but to come up with ideas and models that correctly predict what we can observe. This is called empirical adequacy. For example, in quantum mechanics (the science of very tiny particles), the math can predict things very well, but different scientists have totally different beliefs about what is really happening. Some realists believe in things like the Many-worlds Interpretation, while anti-realists prefer ideas like the Copenhagen Interpretation, which focus only on what we can measure and see. Another big idea in the philosophy of science is how science explains things. One early idea was the deductive-nomological model, created by Carl Hempel. He said that to explain something scientifically, you had to show how it follows from general laws and starting conditions. For example, explaining why a planet moves in an ellipse (oval shape) using Newton's laws of gravity and motion. But not everyone

agreed with this model. Some philosophers thought it did not explain things in everyday life or in complicated sciences like biology. So, newer thinkers, like Nancy Cartwright and James Woodward, introduced different ways to explain things, especially using causes and mechanisms. In this newer view, a good explanation tells us how something works step by step. For instance, if we want to explain how the heart pumps blood, we do not just give a law, we show how parts like pacemaker cells, ion channels, and muscle fibers all work together to make the heart beat. This kind of explanation is especially useful in subjects like neuroscience and medicine, where understanding the parts and how they interact is key.

One big question in the philosophy of science is called the problem of induction. This idea was first brought up by a philosopher named David Hume in the 1700s. He asked: How do we really know that what happened in the past will keep happening in the future? For example, we believe the sun will rise tomorrow because it has always done so. But Hume pointed out that we do not have a perfect reason to believe that, just past experience. Science often depends on these kinds of patterns, but there is no absolute proof that they will always hold true. Later on, a philosopher named Nelson Goodman added to this puzzle. He showed that it is not always easy to decide which patterns are meaningful. He created a strange example using the color “grue,” which means something is green before a certain time and blue after. This example showed that choosing which patterns to trust is harder than it seems. Today, some philosophers and scientists try to solve this problem using probability. This is called Bayesian reasoning, where people adjust their beliefs based on how likely something is and what they already know. But even this method has challenges, like being hard to calculate and including some personal judgment. Another important thinker was Thomas Kuhn, who wrote a famous book in 1962 called *The Structure of Scientific Revolutions*. He said that science does not always grow in a straight line by slowly adding new facts. Instead, he said science works through paradigms, big ideas or frameworks that shape how scientists think, what they study, and what evidence they trust. Scientists work within a paradigm for a long time during “normal science,” but eventually they find anomalies, things that do not fit. When enough of these build up, a scientific revolution happens, and the old paradigm is replaced with a new one. For example, when scientists moved from Newton’s laws to Einstein’s theory of relativity, or from Ptolemy’s geocentric model (Earth at the center) to Copernicus’s heliocentric model (Sun at the center), these were paradigm shifts. Kuhn also said that different paradigms can be incommensurable, meaning they are so different that they cannot easily be compared. This idea was controversial. Philosophers like Karl Popper and Imre Lakatos disagreed with Kuhn. Karl Popper, for example, believed that science moves forward by falsification. This means that scientists should constantly test their ideas and try to prove them wrong. If a theory fails the test, it should be rejected. Popper thought that a good scientific theory must be one that could be

proven false if the evidence does not support it. For example, saying “All swans are white” is a scientific idea because it can be tested, and even disproven by finding just one black swan. Another philosopher, Imre Lakatos, offered a different view. He did not think science was as dramatic as Kuhn’s idea of sudden “paradigm shifts,” but he also did not fully agree with Popper. Lakatos said that science works through research programmes, groups of theories and ideas that develop over time. These programmes may face problems and make mistakes, but if they keep improving and helping us understand more, then they are still making progress. Lakatos believed science should still be rational and goal-directed, even if it changes slowly and builds on earlier ideas. The philosophy of science does not just focus on how we know things or what reality is made of. It also asks important questions about ethics (what is right or wrong) and politics (how science affects society). These questions are especially important in fields like environmental science, artificial intelligence, biotechnology, and how we respond to pandemics. For example, when scientists or governments use computer models to make decisions about public health or climate change, philosophers ask: Are these models reliable? Are they being used fairly? Scientists also have to think about responsibility, how their discoveries might help or harm people. One idea that comes up is called the precautionary principle. This means that if a new technology might be dangerous and we do not fully understand the risks, we should be careful and not rush into using it. This idea is based on epistemic humility, which means recognizing the limits of our knowledge. A good example is CRISPR, a powerful gene-editing tool. While it can be used to treat diseases, it also raises big questions: Should we change human DNA? Should we allow “designer babies”? These are not just scientific questions, they are philosophical and ethical ones, too.

Philosophers also talk about whether there is only one correct way to do science. Some thinkers, like Paul Feyerabend, believed that there is no single method that all science must follow. He called this idea epistemological anarchism, which means that even unusual or untraditional approaches can lead to progress. This way of thinking is called pluralism, and it accepts that different fields of science may need different tools and ways of thinking. For instance, in climate science, scientists use a mix of physics, statistics, and information about human behavior to make predictions. In medicine, both controlled experiments and patient stories give useful, but very different, kinds of evidence. This shows that science does not always follow just one path. Instead, different methods can work together to give us a better understanding of the world. Imagine we are going on an amazing adventure to learn about this! Scientific discoveries have played a big role in changing the world. They have helped improve how people live, work, and communicate. Some important examples include the invention of the printing press, the rise of factories during the Industrial Revolution, and modern tools like electricity, antibiotics, and the internet. These discoveries changed not only technology, but also how

societies are organized. For example, in the 1700s, James Watt improved the steam engine. This invention made it easier to move goods and power machines. It helped build large cities and connect countries through trade. In 1928, Alexander Fleming discovered penicillin, the first antibiotic. This medicine saved many lives by treating infections and changed the way doctors and hospitals work all over the world. Science also helps governments make better decisions. When leaders use evidence and facts, they can solve big problems like climate change, diseases, and food shortages. For example, the Intergovernmental Panel on Climate Change (IPCC) studies thousands of scientific papers to give advice about global warming. In 1987, the Montreal Protocol helped protect the ozone layer by banning harmful gases called CFCs. Scientists showed that CFCs were damaging the atmosphere, and the agreement helped the ozone layer begin to heal. During the COVID-19 pandemic, groups like the World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) gave advice based on science. They told people how to stay safe using masks, vaccines, and social distancing. However, not everyone followed or trusted this advice equally. Science also helps countries work together. This is called science diplomacy. Projects like the International Space Station (ISS) or joint research in the Arctic bring scientists from different countries together. These efforts can build peace and cooperation between nations.

The connection between science and society goes both ways. Science changes the world, but society also shapes how science is done. People's values, culture, money, and politics all influence what research is supported and how results are used. For example, during the Cold War, the U.S. focused heavily on space research after the Soviet Union launched Sputnik 1 in 1957. This led to the creation of NASA and more investment in science and engineering education. Politics and fear of falling behind in space technology drove this big push. Social movements can also change science. In the 1970s, feminist groups pointed out that medical research mostly focused on men. Because of this, rules changed so women had to be included in clinical trials. This helped scientists understand how treatments can work differently in men and women. In farming, many people worry about genetically modified organisms (GMOs). This public concern has led to different rules in different places. For example, Europe uses the precautionary principle, which means being more careful about approving new GMO products than in countries like the U.S. Even the way scientific papers are shared is shaped by money and business models. Many journals charge people to read studies, which has caused debates about whether scientific knowledge should be free and open to everyone. Science also helps grow economies and create new jobs. Many big industries, like biotechnology, renewable energy, artificial intelligence, and aerospace, are based on years of scientific research. For example, the semiconductor industry, which makes computer chips, came from early work in solid-state physics. Today, it is a global industry worth trillions of dollars. Governments often help science and

technology grow by giving tax breaks, supporting patents, and creating partnerships between public and private groups. In countries like South Korea and Israel, spending more than 4% of the country's income on research and development (R&D) has helped build strong technology industries. However, new technology can also cause problems. Sometimes, machines and computers take over jobs that people used to do. This is called technological unemployment. It raises hard questions about what to do when people lose their jobs because of automation. Ideas like universal basic income, job training programs, and better education are being discussed to help people adapt to these changes. Understanding science is important for everyone. It helps people make smart choices, stay healthy, and take part in important debates. This is called scientific literacy. When people understand basic science, they can check if information is true and avoid being misled. If many people do not understand science, it can lead to misinformation. For example, some people still believe vaccines cause autism. This false idea came from a study that was later proven wrong and taken down, but the anti-vaccine movement still exists and causes real harm. Other examples include denying climate change or following pseudoscientific health ideas that are not supported by evidence. To help the public understand science better, many things are being done. Schools are improving science classes, scientists give public talks, and science journalists write easy-to-understand articles. Organizations like the American Association for the Advancement of Science (AAAS) and the British Science Association also organize science festivals and help scientists learn how to talk clearly to the public. People can also take part in citizen science. These are projects where regular people help collect data for scientific studies. Examples include watching birds through eBird or testing water through FreshWater Watch. These projects help people feel more connected to science and show that science is for everyone, not just experts.

Chapter 9

As science becomes more powerful, it also raises ethical and philosophical questions. New technologies like gene editing, brain-computer links, and geoengineering can change human life and the planet in big ways. People must think carefully about what is right or wrong when using these tools. For example, a tool called CRISPR-Cas9 might one day cure genetic diseases. But it could also be used to create “designer babies,” which may increase unfairness in society or bring back dangerous ideas like eugenics. Artificial intelligence (AI) is another case. When AI is used to help police or choose job applicants, it can carry hidden biases from the data it was trained on. This has led to calls for fair and transparent AI systems. To deal with these big issues, there are special groups that work

on science and ethics. These include The Hastings Center, the Nuffield Council on Bioethics, and UNESCO's Bioethics Program. They help create rules and ideas that respect human rights, justice, and the limits of science. These discussions are not just for scientists. They also involve ethicists, lawyers, religious leaders, and the general public. This helps make sure that science is used in ways that are fair and good for everyone. Many of today's biggest problems, like diseases, climate change, and food shortages, affect the whole world. Solving them requires international cooperation and fair access to scientific knowledge. But there are still big differences between richer and poorer countries, often called the Global North and the Global South. Wealthy countries usually lead in science. They have more money for research, better equipment, and more universities. In contrast, many low- and middle-income countries struggle with problems like not enough funding, brain drain (when scientists leave to work in richer countries), and limited access to scientific tools and journals. Some groups are working to close this gap. For example, TWAS (The World Academy of Sciences) and INASP (International Network for the Availability of Scientific Publications) help support science in developing countries. They promote regional cooperation, open-access publishing, and training programs to build research skills. The idea of open science, supported by UNESCO, says that scientific knowledge should be freely available to everyone. It also encourages scientists to work together, share results, and be transparent about how they do their research. A good example of open science is what happened during the COVID-19 pandemic. Scientists around the world used platforms like GISAID to quickly share data about the virus's genes. This helped track new versions of the virus and develop better responses faster. However, there are still challenges. Sometimes, patents and intellectual property rules make it hard for poorer countries to get important technologies, like vaccines or medical treatments. This shows that fairness in science is not just about sharing knowledge, it is also about making sure everyone can use it. Close your eyes and picture this... Science is the best tool humans have for understanding and predicting how nature works. But it is not perfect. Science has limits, and it is not always free from mistakes or bias. Science depends on observation, testing, and the idea that ideas should be falsifiable, which means they can be proven wrong if new evidence shows up. Because of this, science cannot answer questions that are metaphysical (about things beyond the physical world), emotional, or based on personal values. These kinds of questions cannot be tested or measured with tools and experiments. For example, science can study what happens in the brain during meditation, but it cannot fully explain what a person feels during that experience. This is part of a big question in philosophy and brain science called the problem of consciousness. Scientists also disagree about how best to explain complex systems. Some believe that everything can be broken down into smaller parts (reductionism), while others think that new properties appear when parts come together (emergentism). This is an important debate in areas like neuroscience and systems biology.

This is one of my favorite things to teach! Some philosophers have said that science is not as objective as it may seem. For example, Thomas Kuhn argued that what scientists observe is affected by the theories they already believe. This idea is called theory-ladenness. In his famous book *The Structure of Scientific Revolutions* (1962), Kuhn said that science does not always move forward step by step. Instead, it goes through big changes called paradigm shifts, times when one way of thinking is replaced by another. A good example is how Einstein's relativity replaced Newton's mechanics. Kuhn also said that during these shifts, scientists working under different theories may not agree on what counts as proof or even what questions are important. This makes it hard to compare old and new ideas fairly. Another thinker, Paul Feyerabend, said that there is no single "scientific method" that always works. In his book *Against Method* (1975), he said that in the history of science, sometimes "anything goes." He believed that too many strict rules in science can actually stop progress. One example is how Galileo's telescope was at first seen as untrustworthy by many scientists of his time. People did not trust what it showed because it was new and did not fit their existing ideas. This shows that new tools and ideas are sometimes rejected, not because they are wrong, but because they challenge the way people already think. One basic problem in science is called the problem of induction. It was explained by philosopher David Hume in the 1700s. The problem asks: how can we be sure that something will always happen just because it has happened many times before? For example, the sun has risen every day in history, but that does not guarantee it will rise tomorrow. Yet, science often assumes that nature follows regular patterns like this. Philosopher Karl Popper tried to solve this problem by saying that science should focus on proving ideas false, not just confirming them. This is called falsifiability. According to Popper, science moves forward by getting rid of wrong theories. But this approach also has problems. If an experiment gives unexpected results, scientists do not always throw out the theory. Sometimes they blame the tools, the way the data was read, or unknown factors. For example, if a telescope shows strange data about a planet, the issue might be with the telescope, not the theory. So it is not always clear when a theory has truly been disproven. This shows that scientific ideas are not final truths. They can change when new evidence appears or when old evidence is understood in a new way. Social and cultural factors also affect science. They influence what gets studied, who gets to do the research, and how results are shared. Most science today is led by institutions in the Western world, and it is often written in English. This has raised concerns about epistemic colonialism, where knowledge from non-Western cultures, like Indigenous traditions or ancient medical systems, is ignored or only accepted when explained in Western scientific terms. For example, Ayurveda and Traditional Chinese medicine have been used for thousands of years. But in modern science, they are often not taken seriously unless specific parts of them are tested in lab conditions or clinical trials. There are also problems inside the scientific community. Many researchers are under pressure to publish lots of papers to keep their jobs. This is

called "publish or perish." It can lead to rushed studies, weak results, and even bad practices like p-hacking (changing data to get the desired result). These issues are part of what's known as the replication crisis, when other scientists try to repeat a study but do not get the same results. One big study in 2015, called the Open Science Collaboration, tried to repeat 100 famous psychology experiments. Only 36% of them gave similar results. This showed that many scientific findings may not be as reliable as once thought.

Science works best when it is honest and independent. But sometimes, economic and political pressures can affect what scientists study, how results are shared, and whether the truth is told. For example, in the 1900s, tobacco companies paid for studies that made smoking look safer than it really was. They tried to hide the fact that smoking causes cancer. This delayed laws to protect people and led to millions of deaths. The fossil fuel industry has done something similar. By funding certain research groups and articles, they tried to make people doubt the strong scientific agreement that human actions are causing climate change. These actions are examples of corporate influence over science, which can reduce public trust and keep people from knowing the full truth. Governments can also interfere with science. During the early COVID-19 pandemic, some leaders delayed sharing important data or ignored the advice of health experts. This may have made the pandemic worse and hurt people's trust in science and government decisions. To keep science honest, there must be transparency (open sharing of methods and results) and safeguards that protect scientists from outside pressure. Science often uses controlled experiments to find clear answers. But some problems are too complex for this method to work perfectly. Fields like climate science, epidemiology (the study of diseases), and ecology deal with systems that have many parts working together. Because of this, scientists must use models and probabilities to make predictions. For example, climate models look at how things like greenhouse gases, ocean currents, and weather patterns interact. These models can show possible future trends, but they cannot predict exactly what will happen in a specific place at a specific time many years ahead. Sometimes people misunderstand this. They think that if science is uncertain, it must be wrong. But in complex systems, uncertainty is normal. It does not mean the science is bad. The same problem appears in nutritional science. It is hard to study diet because people eat many things at once, and other factors (like sleep or exercise) affect health too. This can lead to confusing or changing advice about what is healthy. Even in areas with experiments, like genetics, scientists have learned that things are more complicated than they once thought. Many diseases cannot be explained by a single gene. Instead, they are caused by interactions between many genes and environmental factors. This shows that reductionism, studying only small parts of a problem, can miss the bigger picture when dealing with complex systems. Here's a really cool secret about this topic: Science is a powerful way to learn about the natural world, but it has limits. Some important questions

cannot be answered by science alone. These include normative (what we should do), existential (why we exist), and metaphysical (beyond the physical world) questions. For example, science cannot fully answer “What is the meaning of life?”, “Is there a God?”, “What is morally right or wrong?”, or “What is consciousness?”. Science can study things we can see and test, like how the brain works. For instance, neuroscience can show how parts of the brain light up when we make choices. But it cannot tell us what we should do morally. This is known as the is–ought problem, explained by philosopher David Hume. In the same way, evolutionary biology can explain how helping others (altruism) may have developed in humans. But it cannot say whether we should be kind from a moral point of view. These types of questions are explored through philosophy, religion, ethics, and the humanities (like history and literature). Thinking that science is the only way to gain knowledge is called the scientism fallacy. It ignores the important insights from other areas of human thought, like art, culture, and spiritual traditions.

People’s trust in science is not the same everywhere. It depends on their background, politics, and past experiences. In some places, people see science as something only for experts or something that does not care about their real-life problems. This is especially true in communities that have been treated unfairly by scientific institutions in the past. One well-known case is that of Henrietta Lacks, an African-American woman whose cancer cells were taken without her permission in the 1950s. Her cells were used in many scientific breakthroughs, but her family was never asked or informed. This case is still discussed today in conversations about bioethics (the ethics of medical and scientific research) and trust in science. To build trust, it is not enough to just explain science better. Scientists also need to include different communities, listen to their concerns, and make research more open and fair. People also get confused when scientific advice changes. For example, during the COVID-19 pandemic, guidelines about wearing masks changed as scientists learned more. Some people saw this as a sign that science was unreliable. But in reality, this shows how science works, it updates when new evidence becomes available. That is a strength, not a weakness, but it must be clearly explained so the public understands why advice sometimes changes. "How Do We Know What Is True?" (animated video; 2:52) Biology is the science that studies life, living things, and the evolution of life. Living things include animals, plants, fungi (such as mushrooms), and microorganisms such as bacteria and archaea.

The term 'biology' is relatively modern. It was introduced in 1799 by a physician, Thomas Beddoes. It comes from the Greek word βίος (bios), "life", and the suffix -λογία (logia), "study of". Did you know? People who study biology are called biologists. Biology looks at how animals and other living things behave and work, and what they are like. Biology also studies how organisms react with each other and the

environment. It has existed as a science for about 200 years, and before that it was called "natural history". Biology has many research fields and branches. Like all sciences, biology uses the scientific method. This means that biologists must be able to show evidence for their ideas and that other biologists must be able to test the ideas for themselves. Imagine we are going on an amazing adventure to learn about this! "What are the characteristics of this living thing?" (comparative anatomy)

Chapter 10

This is one of my favorite things to teach! "How should we group living things?" (classification, taxonomy) "What does this living thing do?" (behaviour, growth) "How do living things relate to their environment?" (ecology)

Did you know? Modern biology is influenced by evolution, which answers the question: "How has the living world come to be as it is?" Did you know? The earliest roots of science, including medicine, go back to ancient Egypt and Mesopotamia around 3000 to 1200 BCE. Their ideas influenced ancient Greek natural philosophy. Greek thinkers like Aristotle (384–322 BCE) added a lot to early biology. He studied causes in nature and the variety of living things. His student, Theophrastus, began the scientific study of plants. Here's a really cool secret about this topic: In the medieval Islamic world, several scholars wrote about biology. Al-Jahiz (781–869) and Al-Dīnawarī (828–896) wrote about animals and plants. Rhazes (865–925) studied the body and how it works. Islamic scholars, working from Greek ideas, studied medicine deeply. They also followed Aristotle's ideas about nature.

Let me tell you a fascinating story. Biology grew quickly when Anton van Leeuwenhoek greatly improved the microscope. Scientists then discovered sperm cells, bacteria, and many tiny living things. Jan Swammerdam's work led to interest in insects and new methods of studying tiny organisms. Better microscopes strongly changed how people thought about life. In the early 1800s, scientists saw that cells were central to life. In 1838, Schleiden and Schwann said that (1) all living things are made of cells and (2) each cell shows the signs of life. They did not believe that (3) all cells come from other cells; they still thought cells could appear on their own. Later, Robert Remak and Rudolf Virchow proved that new cells come from existing cells. By the 1860s, most scientists accepted all three ideas, which became known as cell theory. At the same time, natural historians focused on naming and grouping living things. Carl Linnaeus published a basic system in 1735 and gave scientific names to species in the 1750s. Georges-Louis Leclerc, Comte de Buffon, believed

species were not fixed and could change, and even suggested that different species might share ancestors.

Imagine we are going on an amazing adventure to learn about this! In 1842, Charles Darwin wrote the first draft of *On the Origin of Species*. Earlier, Jean-Baptiste Lamarck had offered one of the first full theories of evolution. Darwin built a stronger theory of evolution through natural selection. He used ideas from geography, geology, and population studies, along with his own observations. Alfred Russel Wallace reached similar conclusions on his own. This is one of my favorite things to teach! Modern genetics started with Gregor Mendel's work in 1865. He explained how traits are passed from parents to offspring. People did not fully understand his work until the early 1900s. Then, evolution and genetics were combined into the modern synthesis. In the 1940s and early 1950s, experiments by Alfred Hershey and Martha Chase showed that DNA is the part of chromosomes that carries genes. Scientists began to study small organisms like viruses and bacteria. In 1953, James Watson and Francis Crick discovered that DNA has a double-helix shape. After this, biology expanded into molecular genetics. Scientists cracked the genetic code, showing how DNA uses codons to build proteins. This work was done by Har Gobind Khorana, Robert W. Holley, and Marshall Warren Nirenberg. In 1990, the Human Genome Project began to map all human genes.

All living things are made of chemical elements. Most of their mass (about 96%) comes from oxygen, carbon, hydrogen, and nitrogen. Other elements like calcium, phosphorus, sulfur, sodium, chlorine, and magnesium make up almost all the rest. These elements can join together to form compounds, such as water, which is essential for life. Biochemistry is the study of chemical processes in living things. Molecular biology is the study of how molecules in cells work, including how they are made, changed, and how they interact with each other. This is one of my favorite things to teach! Life began in Earth's first oceans about 3.8 billion years ago. Since then, water has been the most common molecule in all living things. Water is important for life because it is a great solvent. This means it can dissolve many substances, like salt (sodium and chloride ions) and small molecules, to form solutions. When substances are dissolved in water, they can more easily meet and react with each other, which is necessary for life.

Chapter 11

Because of this polarity, water molecules attract each other using hydrogen bonds, which makes water cohesive (they stick together). You did an incredible job! We learned so much today. I'm very proud of

you!

